

Practical -4.

Aim:

Setup Configure a LAN (local area network) using a switch and ethernet cables in your lab.

How to Setup a LAN:

Step 1: Plan and Design an appropriate network topology taking into account network requirement and equipment location.

Step 2: Connect your Computers to network switch via an Ethernet cable, which is as simple as plugging one end of the Ethernet into your computer and the other end into your network switch.

Step 3: Connect your Computers to network switch via an Ethernet cable, which is as simple as plugging one end in your network switch.

Step 4: Assign IP address to your PCs.

1. Log on to the client computer as administrator or as owner.

2. Click network and Internet connection.

3. Right click local Area Connection / Ethernet → Go to Properties → Select Internet Protocol (TCP/IPv4) → Click

on properties → Select use the following IP address.

Step 5 - Configure a network switch.

1. Connect to your computer to switch.

2. Login in to user interface.

3. Configure basic settings.

4. Assign IP address as 10.1.1.0 Subnet mask 255.0.0.0

Step 6: Check the connectivity between switch and other machines by using Ping command in the command prompt of device.

Step 7: Select a folder → go to properties → Click sharing tab → Share with everyone on same LAN.

Step 8: Try to access the shared folder from other computers of network.

Student observation:

Write the configuration of each and every device, write the outcome and challenges faced while configuring LAN.

1. LAN was setup successfully.

2. A device like computers and printers were connected and could share files.

3. Internet was working on all devices.

4. IP address were set correctly.
5. The network was tested and worked well.
Challenges faced while configuring LAN:

1. Some devices had the same IP address causing connection problem.
2. Some cables were faulty are not connected properly.
3. Settings on routers or switch were sometimes wrong.

Result:

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Thus a LAN is setup and configured.